BONDU CHANDU

Hyderabad, Telangana, India | +91 6303461146 | chanduchowdary8978@gmail.com
LinkedIn | GitHub | Portfolio

Currently Working On

Ride-Hailing Platform Simulation (RapidoSim) Apr 2026 – Present

- Simulating a Rapido-style bike-taxi platform across 24 spatial zones modeled on Hyderabad's urban market, with multi-agent supply-demand dynamics and zone-level spatial coupling
- Implemented bandit/RL pricing policies (UCB1, Thompson Sampling, Q-learning); driver agent modeled as optimizer using Bellman TD learning
- Integrating variance-reduced gradient methods (SVRG, SPIDER) into the optimization layer; modeling endogenous latency and fatigue-driven supply erosion as dynamic state variables

Professional Summary

ML Engineer with strong foundations in convex optimization, statistical modeling, and stochastic simulation. Experienced in building end-to-end ML pipelines, time-series forecasting, and simulation-driven decision systems. Focused on reliability, uncertainty estimation, and principled optimization — spanning both research-grade mathematical analysis and applied system-level implementation.

Technical Skills

- **Programming:** Python, R, C, C++, SQL
- **Machine Learning:** Regression, Classification, Feature Engineering, Model Evaluation, Ensemble Methods, Calibration, Uncertainty Estimation
- **Deep Learning:** LSTM, Neural Networks
- **Libraries:** NumPy, pandas, scikit-learn, PyTorch, LightGBM, SHAP, Matplotlib, Seaborn, PySpark
- **Optimization:** Gradient Descent, Convex Optimization, KKT Conditions, Proximal Methods, Variance Reduction (SVRG, SPIDER), Lipschitz Smoothness Analysis
- **Tools & Systems:** Git, Linux, Jupyter Notebook, Streamlit, Docker, MLflow, MongoDB, PostgreSQL, Kafka

Education

M.Tech — Mathematics and Computing Aug 2025 – May 2027 (Expected)
Dr. B. R. Ambedkar National Institute of Technology, Jalandhar CGPA: 7.77

B.Tech — Computer Science and Engineering Aug 2020 – May 2024
JNTUA College of Engineering, Kalikiri CGPA: 7.66

Experience

Machine Learning Intern Feb 2024 – May 2024
Indian Space Research Organisation (ISRO) – SDSC SHAR, Andhra Pradesh, India

- Developed LSTM-based time-series forecasting model on real operational weather datasets in a production research environment
- Built end-to-end training and evaluation pipeline in PyTorch for multi-step sequence modeling across variable forecast horizons
- Analyzed prediction errors across multi-horizon forecasts to evaluate generalization, temporal stability, and model degradation patterns
- Studied model robustness under real-world constrained data conditions; performed error decomposition to isolate sources of forecast drift

Technical Author — Learning Optimization Newsletter Link Jan 2026 – Present

- Publishing a structured series on optimization algorithms with mathematical derivations and from-scratch Python implementations; topics include variance-reduced methods (SVRG, SPIDER), momentum, adaptive learning rates, L-BFGS curvature approximation, and natural gradient methods (K-FAC)

Projects

Uncertainty-Aware Cardiovascular Risk Decision System GitHub Jan 2026 – Mar 2026

- Benchmarked 5 models (Logistic Regression, KNN, Random Forest, Gradient Boosting, LightGBM); best model achieved ∼0.80 AUC on held-out test set
- Engineered domain-informed features including BMI, pulse pressure, and mean arterial pressure to improve discriminative signal
- Incorporated cost-sensitive decision thresholds to maximize recall of high-risk patients under asymmetric loss
- Estimated prediction uncertainty via bootstrap sampling; mapped outputs to a 4-tier risk system (High Risk, Moderate, Low Risk, Uncertain)
- Applied SHAP for global and local explainability; performed probability calibration using reliability curves and Brier score; tested robustness under noise and distribution shift

Thermal Throttling Prediction using System Telemetry GitHub Sep 2025 – Nov 2025

- Formulated a dual-task ML problem: binary classification (throttle event detection) and regression (temperature estimation) using real HWiNFO telemetry logs
- Demonstrated temporal features significantly outperform static snapshots; achieved sub-1°C average error in temperature regression
- Engineered time-aware rolling-window features enabling early anomaly detection before observable performance degradation
- Designed the system as a predictive early-warning pipeline; validated applicability in performance-constrained and edge deployment settings

Convex Optimization Essentials for Linear Models GitHub Nov 2025 – Jan 2026

- Implemented Batch GD, SGD, and Mini-batch gradient descent from scratch for linear and logistic regression in NumPy and PyTorch
- Analyzed Lipschitz smoothness, conditioning, and convergence behavior; studied how learning rates and data geometry affect optimization trajectories
- Derived and verified KKT conditions, subgradients, and regularization geometry (L1/L2); visualized loss landscapes to build geometric intuition
- Evaluated convergence rates and generalization on real datasets with ablations over learning rate schedules and batch size regimes

Hyperparameter Optimization Dashboard GitHub Nov 2023 – Jan 2024

- Built an interactive Streamlit application for dataset upload, model selection, and automated hyperparameter tuning
- Supported grid search and random search across multiple ML algorithms with configurable parameter spaces
- Visualized tuning trajectories, convergence curves, and final performance metrics; designed for direct model-to-model comparison

Titanic Survival Prediction GitHub 2023

- Compared multiple classification models with systematic feature engineering; applied ensemble methods to improve out-of-sample generalization
- Built structured evaluation framework for cross-model performance analysis; analyzed feature importance to identify key predictors

Research

Manuscript in Preparation — Theoretical study of asymptotic behavior in spatial queuing systems under stochastic supply-demand dynamics. Contributions include fluid and diffusion limit theorems, endogenous latency modeling, Lyapunov stability analysis, throughput ceiling characterization, and large deviations bounds. Targeting *Queueing Systems* / *Operations Research Letters* (Expected Submission: Jul 2026)